Calorific value of different fuels

Fuel	Calorific Value (kJ / kg)
Cow dung cake	6000-8000
Wood	17000-22000
Coal	25000-33000
Petrol	45000
Kerosene	45000
Diesel	450000
Methane	500000
CNG	50000
LPG	55000
Biogas	35000-40000
Hydrogen	150000

The amount of heat energy produced on complete combustion of 1kg of fuel is called its calorific value. The calorific value of a fuel is expressed in a unit called kilo joule per kg (kJ/kg)

Calorific value = Heat produced / Amount of fuel used for burning in kJ/kg

If 4.5kg of fuel is completely burnt and the amount of heat produced stands measured at 1, 80,000 kJ what is its calorific value.

Calorific value = 1, 80,000 / 4.5 = 40,000 KJ/Kg

Types of combustion

There are three main types of combustion. They are,

Rapid combustion: It is a combustion process in which a substance burns rapidly and produces heat and light with the help of external heat. E.g. Burning of LPG.

Spontaneous combustion: Is combustion process in which a light without the help of external heat.eg. Phosphorus burns spontaneously at room temperature.

Explosion: It is a type of combustion in which a substance burns suddenly and produces heat, light and sound with the help of heat or pressure. E.g. Explosion of crackers.

More Environmental Impacts

CO- Leads to respiratory problem

CO_2- Global warming

SO_2/NO_2 – Acid Rain







Remedies





VII STD Science Term-3 EM unit 4.indd 75 09-11-2019 12:47:19